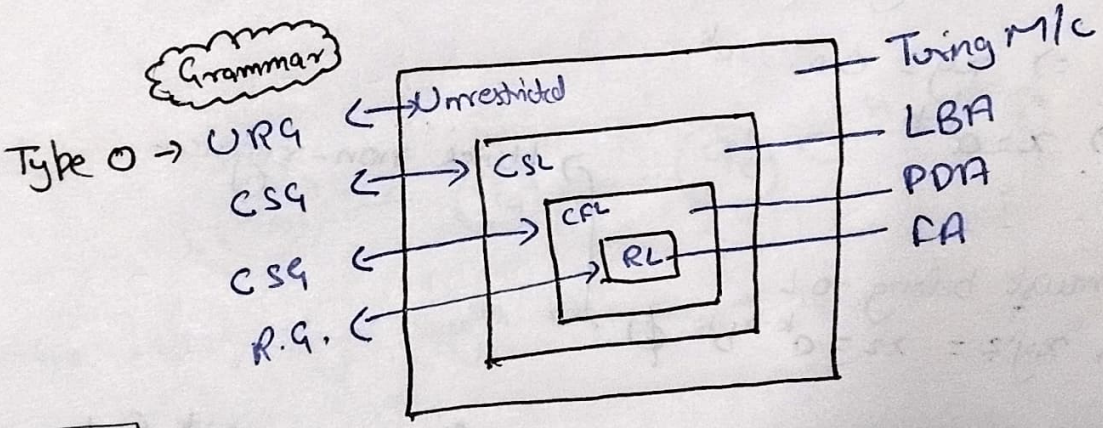
Hence, non-regu

Sessional - 1 Syllabus Complete

22 sept, 2026

Chomsky Heirarchy



Variables

Sentence → <Pronoun> <Verb> <Adjective> .

<Pronoun> → He/She

<Verb> → Walks/runs

<Adjective> → Slowly / Quickly } → P

Terminals

Wednesday
23 Sept, 2026

$G = \langle V, \Sigma, P, S \rangle$

$V \rightarrow$ FNEs of variables

$\Sigma \rightarrow$ FNEs of terminals

$P \rightarrow$ FNEs of rules/productions

$\$ \rightarrow$ Start symbol $[\$ \in V]$

$P: \{ \; \$ \Rightarrow PVA.$
$P \rightarrow h/s$
$V \rightarrow w/a$
$A \rightarrow s/q \; \}$

Previous ex.:

$V = \{ \langle sent \rangle, \langle pronoun \rangle, \langle verb \rangle, \langle Adj \rangle \}$

$\Sigma = \{ He, she, walks, owns, quickly, slowly, \$ \}$ ($\$ \rightarrow$ full stop (terminal))

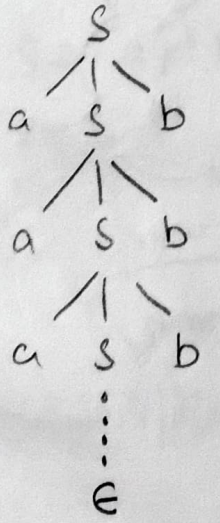
$\$ = \langle sentence \rangle$

$\Rightarrow \$ \rightarrow a\$/\epsilon$

$\rightarrow S$ $\rightarrow S$
$\rightarrow aS$ $\rightarrow \epsilon$
$\rightarrow a.\epsilon$
$\rightarrow a$.

$S \rightarrow aS/\epsilon$
↳ Repetitive S [R.G.]

$S \rightarrow aSb/\epsilon$
↳ Recursive S [C.F.G.]



(LMD)
(Left-most derivation)

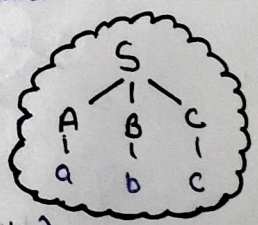
$S \rightarrow ABC$
$A \rightarrow a$
$B \rightarrow b$
$C \rightarrow c$

$\Rightarrow S$
$\rightarrow ABC$
$\rightarrow aBc$
$\rightarrow abC$
$\rightarrow abc$
↳ Sentence.

Derivation.
sentential form
Level ↑

$\rightarrow S$
$\rightarrow ABC$
$\rightarrow ABc$
$\rightarrow Abc$
$\rightarrow abc$.

(Right-most derivation)
(RMD)



Monday
8 Sept, 2018

Type 0 : $\alpha \rightarrow \beta$ $\quad \alpha, \beta \in (V_n \cup \Sigma)^*$ & $\alpha \neq \epsilon$

$aBcD \rightarrow xYz$

Type 1 : $\alpha A \beta \rightarrow \alpha \lambda \beta$ $\quad \lambda, \alpha, \beta \in (V_n \cup \Sigma)^*$ & $A \in V_n$

$abCde \rightarrow abDCde$

$\alpha \rightarrow$ left context

$\beta \rightarrow$ Right context

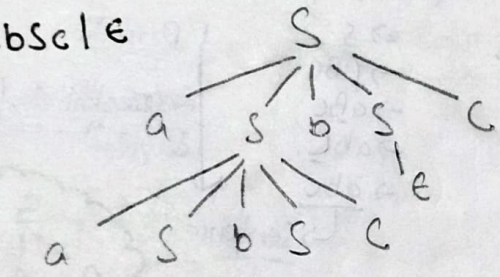
Type 2 : $A \rightarrow \lambda$ $\quad \lambda \in (V_n \cup \Sigma)^*$, $A \in V_n$

$qS \rightarrow aSb|\epsilon$

Type 3 : $A \rightarrow aB|a$ $\quad A, B \in V_n$, $a \in \Sigma_n$

$A \rightarrow Ba|a$

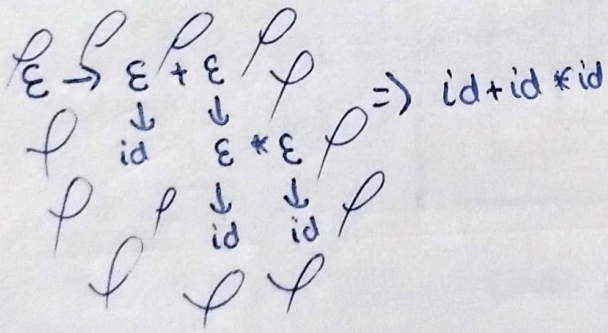
$S \rightarrow aSbSc|\epsilon$



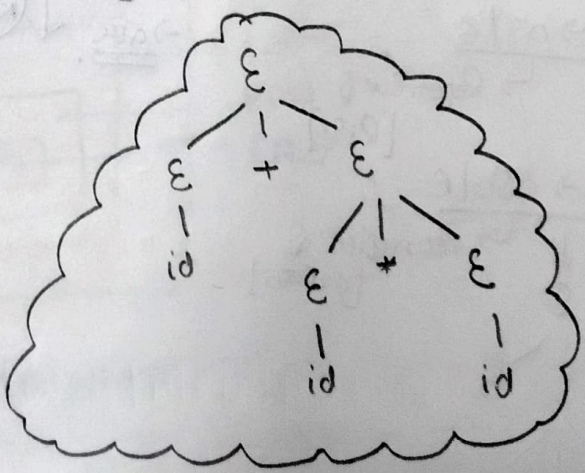
: aabcbc

$\Rightarrow E \rightarrow E+E|E*E|id$ ← single terminal

= id+id*id



$\Rightarrow$ id+id*id



$\rightarrow E$
$\rightarrow E+E$
$\rightarrow id+E$
$\rightarrow id+E*E$
$\rightarrow id+id*E$
$\rightarrow id+id*id$

$\rightarrow E$
$\rightarrow E*E$
$\rightarrow E+E*E$
$\rightarrow id+E*E$
$\rightarrow id+id*E$
$\rightarrow id+id*id$

} Ambiguous grammar
↳ 2 unique LMDs/RMDs/D.Tree

Tuesday
23 Sept 2026

$a^*b^*c^*$ Not required. $\Rightarrow a^* + b^* + c^*$

$\Rightarrow S \to ABC/\epsilon$ $\quad S \to A|B|C$: Chained production (Not good) $\quad S \to aA|bB|cC|\epsilon$

$A \to aA/\epsilon$ $\quad A \to aA|\epsilon$ $\quad A \to aA|\epsilon$

$B \to bB|\epsilon$ $\quad B \to bB|\epsilon$ $\quad B \to bB|\epsilon$

$C \to cC|\epsilon$ $\quad C \to cC|\epsilon$ $\quad C \to cC|\epsilon$ ✓

$\Rightarrow S \to aA|bB|cC|\epsilon$

$A \to aA|bB|cC|\epsilon$

$B \to bB|cC|\epsilon$

$C \to cC|\epsilon$

$\Rightarrow$ aaccc

$\to S$
$\to aA$
$\to aaA$
$\to aacC$
$\to aacCC$
$\to aacccC$
$\to aaccc\epsilon$
$\to aaccc$

$\Rightarrow a^*b^* + c^*d^*$

$\Rightarrow S \to aA|bB|cC|dD|\epsilon$

$A \to aA|bB|\epsilon$

$B \to bB|\epsilon$

$C \to cC|dD|\epsilon$

$D \to dD|\epsilon$

$\Rightarrow (a+b)^*$

$S \to aS|bS|\epsilon$

$\Rightarrow \{a^n b^m / n, m \geq 0 \ \& \ m = n+1\}$

$S \to aSb|b$

Monday
5 Oct, 2026

Context-Free Languages

$V \to (V \cup \Sigma)^*$

$S \to aSb/\epsilon$

Simplification.

$\Rightarrow S \to A$
$A \to B$
$B \to cd$

$\Rightarrow$ $S \to cd$

$\Rightarrow S \to AB$
$A \to a$
$B \to b$
$C \to d$ $\Rightarrow$ X

$S \to AB|CD$
$A \to a$
$B \to b$

$\to S \to AaB$
$A \to a|\epsilon$
$B \to b|\epsilon$

$\Rightarrow$

$S \to AaB|aB|Aa|a$
$A \to a$
$B \to b$
{Better}
{Min. or no 'ϵ'}

4-steps of simplification

1.) Elimination of var. that do not produce terminals
2.) " " " " " " appear in sentential form
3.) " " chained productions (unit productions)
4.) " " null (ϵ) " .

Order of application: $4 > 3 > 2 > 1$

$S \to ABab|Ca$
$A \to a$
$B \to b$
$C \to aCA$

$W_1 = \{X | X \to \Sigma^*\}$

$W_{i+1} = W_i \cup \{X | X \to (W_i \cup \Sigma)^*\}$

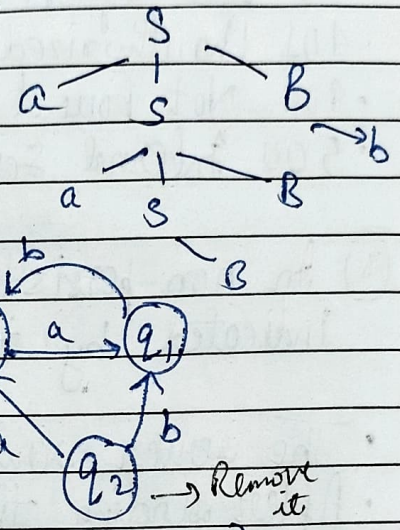
$W_1 = \{A, B\}$
$W_2 = \{A, B\} \cup \{A, B, S\}$
$= \{A, B, S\}$

$\Rightarrow Q' = V' = W_k$
$\Sigma' = \Sigma$
$P' = W_k \to (W_k \cup \Sigma)^*$

7/10/25

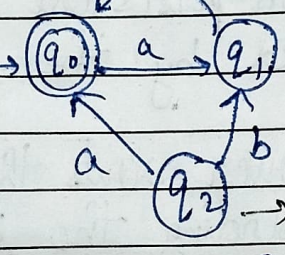
$S \to aSB$ → Useless Grammar/Production
$B \to bB \mid \varepsilon$

But $S \to aSB \mid aB$ Now it is fine
$B \to b \mid \varepsilon$



(2) Sential form, $S \to AB \mid aBb$
$A \to aA \mid \varepsilon$
$B \to bB \mid \varepsilon$
$C \to ab$

Bcz it is not reachable from S.



⇒ $S \to AB$
$A \to aA \mid \varepsilon$
$B \to cD \mid \varepsilon$
$D \to dE$
$E \to e$
$C \to ed$

$W_1 = \{S\}$

$W_{i+1} = W_i \cup \{X \mid W_i \to \alpha \;.\; X \in \alpha\}$
$S \to \alpha$

$W_1 = \{S\}$
$W_2 = \{S\} \cup \{A, B\} = \{S, A, B\}$
$W_3 = \{S, A, B\} \cup \{A, B, \cancel{D, E}, A, cD\}$
$= \{S, A, B, D, a, c\}$
$W_4 = \{S, A, B, D, a, c\} \cup \{\dots, dE\}$
$= \{S, A, B, D, a, c, d, E\}$
$W_5 = \{S, A, B, D, a, c, dE, e\}$

Now, $W_5 = W_6$. Augmentation of G

$G' = \langle V', \Sigma', P', S \rangle$

$V' = W_K \cap V$
$\Sigma' = W_K \cap \Sigma$
$P' = \{V' \to (V' \cup \Sigma')^*\}$

→ Important

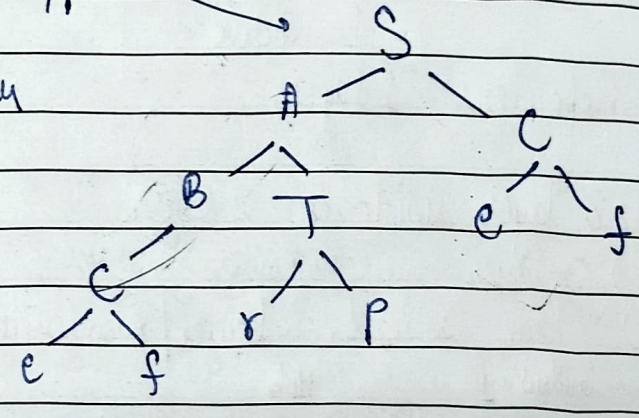
③ Unit Production

$S \to A \mid C$
$A \to B \mid T$
$B \to C$
$C \to e \mid f$
$T \to r \mid p$

$\Rightarrow$

$S \to e \mid f \mid r \mid p$
$A \to e \mid f \mid r \mid p$
$C \to e \mid f$
$T \to r \mid p$

Chain of S, A, B →
Find & compress the chain



∀ V_j in V_n
→ Find the chain
→ Compress the chain

① $W_i(V_j) = \{V_j\}$

$W_{i+1}(V_j) = W_i(V_j) \cup \{X \mid W_i(V_j) \to X\}$, $X \in V_n$

* Pick up all the symbols which comes on RHS from the LHS of the V_j (Previous).

$W_1(S) = \{S\}$ $\quad W_1(A) = \{A\}$
$W_2(S) = \{S\}$ $\quad W_2(A) = \{A\} \cup \{B, T\} = \{A, B, T\}$
$W_3(A) = \{A, B, T\} \cup \{B, T, C\}$
$W_4(A) = \{A, B, T, C\} \cup \{\phi\}$
$= \{A, B, T, C\}$

$V_i \to \alpha \mid \alpha \neq V_n$
$V_j \to$ $\quad A \to \alpha$

2 me chain aayegi {A} & B then compress both.

$\Sigma' = \Sigma$

$P' = \omega_k \to (\omega_k \cup \Sigma)^*$

⋮

Monday
12 Oct, 2026

Elimination of ϵ Productions

$S \to AB \mid AaBb$

$A \to aA \mid \epsilon$

$B \to bB \mid \epsilon$

$\Rightarrow$

$S \to AB \mid AaBb \mid A \mid B \mid \epsilon \mid Aab \mid aBb \mid ab$

$A \to aA \mid a$

$B \to bB \mid b$

$\omega_i = \{X \mid X \to \epsilon\}$

$\omega_{i+1} = \omega_i \cup \{X \mid X \to (\omega_i)^*\}$

Imp:

$\Rightarrow$ E → EaT | ϵ
T → TeA | TA | t
A → ABc | DB
B → CD | b
C → cC | ϵ
D → dD | ϵ

$\Rightarrow$

E → EaT | e
T → TeA | TA | t | Te
A → ABc | DB | Bc | Ac | c | B | D | ϵ (A is not start symbol)
B → CD | b | C | D
C → cC | c
D → dD | d

$\omega_1(A) = \{A\}$
$\omega_2(A) = \{A\} \cup \{B, D\}$
$\omega_3(A) = \{A, B, C, D\}$

$\omega_1 = \{C, D\}$
$\omega_2 = \{B, C, D\}$
$\omega_3 = \{A, B, C, D\}$
$\omega_4 = \{\phi, A, B, C, D\}$
$\omega_5 = \{E, T, A, B, C, D\}$

$\omega_1(B) = \{B\}$
$\omega_2(B) = \{B, C, D\}$

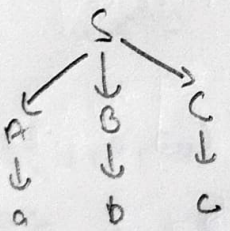
E → EaT | e
T → TeA | TA | t | Te
A → ABc | DB | Bc | Ac | c | CD | b | e | C | d | D | d
B → CD | b | c | C | d | D | d
C → cC | c
D → dD | d

Tuesday
13 Oct, 2026

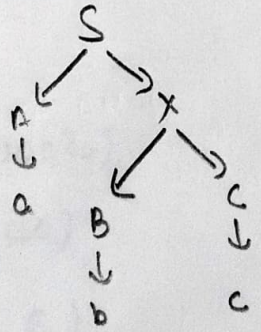
Chomsky Normal Form (CNF)

↳ CFG with binary derived tree

S → ABC
A → a
B → b
C → c

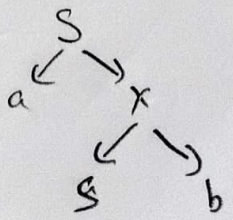


S → AX
X → BC
A → a
B → b
C → c



S → aSb | ϵ

S → aX | ϵ
X → Sb



$S \to ABCD$
$A \to a$
$B \to b$
$C \to c$
$D \to d$

$S \to AP$
$P \to BQ$
$Q \to CD$
$A \to a$
$B \to b$
$C \to c$
$D \to d$

} $n-2$ extra variables where $n = |\text{R.H.S.}|$

(Stair-case approach)

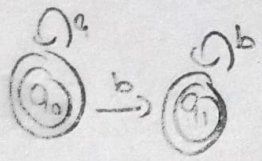
Steps to convert CFG in CNF

1) Simplify the grammar.
2) Replace all combinations of Σ & V on R.H.S. by combination of variables only

$S \to abc \Rightarrow S \to C_a C_b C_c$
$C_a \to a$
$C_b \to b$
$C_c \to c$

3) Reduce the size of R.H.S. of each production to max. 2-variables using stair-case method,

$X \to Y_1 Y_2 Y_3 \dots Y_n \Rightarrow X \to Y_1 C_1$
$C_1 \to Y_2 C_2$
$C_2 \to Y_3 C_3$
$\vdots$
$C_{n-2} \to Y_{n-1} Y_n$



$\Rightarrow \delta(q_0, a) = q_0$
$\delta(q_0, b) = q_1$
$\delta(q_1, b) = q_2$